

Experimental Design Setup for p01.

Based on a thorough review of the provided research paper, here is the critical experimental design setup.

1. Summary of the Paper's Core Method & Evaluation

The paper develops a domain-specific QA system for plant protection by fine-tuning the InternLM-20B model using Low-Rank Adaptation (LoRA) and integrating it with a Retrieval-Augmented Generation (RAG) framework that leverages a specialized vector database. The primary evaluation metrics used to assess the system's performance are ROUGE (for text overlap), BERTScore (for semantic similarity), and FactCC (for factual consistency against a reference).

2. Identified Flaw/Missed Opportunity in Experimental Design

The paper's experimental design fails to include a robust, domain-specific baseline model for comparison. While the proposed system (InternLM-20B-LoRA + RAG) is evaluated using standard NLP metrics, the results are presented in a vacuum. There is no comparison against other powerful, publicly available LLMs—either in their base form, fine-tuned on the same agricultural data, or enhanced with RAG. This omission makes it impossible to determine whether the reported performance gains (e.g., 97% accuracy) are truly due to the novel methodology or if they could be matched or exceeded by existing, more accessible models.

3. Proposed Alternative/Addition (The "AI Scientist's Better Way")

To establish a meaningful and rigorous benchmark, the experimental design must include a comprehensive set of baseline models. The performance of the proposed system should be compared against these baselines on the same test set of agricultural QA pairs.

Specific Baseline Models to Include:

1. General-Purpose State-of-the-Art LLM (Zero-Shot): GPT-4 or a similar top-tier model. This baseline tests whether a powerful, general model without domain-specific fine-tuning can perform well, setting a high bar for the specialized system.
2. General-Purpose LLM with RAG (Few-Shot): The same model (e.g., GPT-4) equipped with access to the paper's own plant protection knowledge base via RAG. This directly isolates the contribution of the custom knowledge base from the contribution of the LoRA fine-tuning.
3. A Fine-Tuned Domain-Specific Competitor: A model of comparable scale (e.g., Llama 3 70B or Qwen 7B) fine-tuned on the same 9,000 QA pairs using full-parameter fine-

tuning or a different PEFT method like QLoRA. This baseline is crucial to demonstrate that the choice of LoRA is not only efficient but also as effective or more effective than other fine-tuning approaches for this specific task.

4. The Base InternLM-20B Model (Without Fine-Tuning or RAG): This serves as an ablation baseline to quantify the individual performance improvements brought by the LoRA fine-tuning and the RAG component.

Implementation of the Comparison:

Each baseline and the proposed model should be evaluated using the paper's chosen metrics (ROUGE, BERTScore, FactCC). Crucially, this should be supplemented by a rigorous human evaluation focused on agricultural experts. Experts should score responses on criteria like factual correctness, actionability of advice, and identification of potential hallucinations on a Likert scale. To ensure the results are statistically robust, a paired T-test or Wilcoxon signed-rank test should be performed on these human evaluation scores to confirm that the performance differences between the proposed system and the key baselines are statistically significant ($p\text{-value} < 0.05$).

This comparative framework would transform the paper's claims from being self-referential to being contextualized within the current AI landscape, providing compelling evidence for the superiority and necessity of the proposed methodology.